Dynamic Programming

Part 2: Exercise 3 (week 13)

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Plan for today

- Policy Iteration as an alternative to VFI
  - Solving Harold Zurcher using Policy Iteration
- The Nested Pseudo Likelihood Algorithm (NPL)
- We will be working in the folder `03_NPL` but calling the model functions from `01_02_NFXP`

Reformulating the integrated value function

We know that Harold Zurcher solves:

$$\bar{V}(x) = \mathbb{E}\left[\max_{d\in\{0,1\}} \exp\left(u(x,d) + \epsilon(d) + \beta P_d^M(x)\bar{V}\right)\right]$$

New thing: def' event $I_d(x)$ that choice d is optimal.

$$\Rightarrow \bar{V}(x) = \sum_{d\in\{0,1\}} P(d|x)\left(u(x,d) + \mathbb{E}[\epsilon(d)|I_d(x)] + \beta P_d^M(x)\bar{V}\right)$$

In matrix form: $$\Rightarrow \bar{V} = \sum_{d\in\{0,1\}} P(d)\left(u(d) + \mathbb{E}[\epsilon(d)|I_d] + \beta P_d^M\bar{V}\right)$$

$$\Leftrightarrow \bar{V}(I - \beta \underbrace{\sum_{d\in\{0,1\}} P(d)P_d^M}_{=F^U(P)}) = \sum_{d\in\{0,1\}} P(d)\left(u(d) + \mathbb{E}[\epsilon(d)|I_d]\right)$$

$$\Leftrightarrow \bar{V} = \left[I - \beta F^U(P)\right]^{-1} \sum_{d\in\{0,1\}} P(d)\left(u(d) + \mathbb{E}[\epsilon(d)|I_d]\right)$$

with $\mathbb{E}[\epsilon(d)|I_d] = \gamma - \ln(P(d))$ when $\epsilon(d) \sim Gumbel(0,1)$

Policy Iteration (PI)

Reformulation of integrated value function $\rightarrow$ mapping: $\bar{V} = \psi(P)$

Other "way" is even easier with logit-form, first scalar, then matrix:

$$P(d=0|x) = \frac{1}{1+\exp(u(x,1)+\beta P_1^M \bar{V} - u(x,0) - \beta P_0^M \bar{V})}$$

$$\Leftrightarrow P = \Lambda(\bar{V})$$

With ψ and Λ, we define the **Policy Iteration Operator** Ψ

$$P = \Lambda(\psi(P)) = \Psi(P)$$

, which inherits its fixed point from the "underlying value function"

$\rightarrow$ Suggests iterating on Ψ, instead of $\bar{\Gamma}$

$\rightarrow$ Remember $\Psi(P)$ is actually function of parameters $\Psi(P;\theta)$

Nested Pseudo Likelihood (NPL) Algorithm

Today: Simplify by excluding state transition in likelihood

Pre-step: Make initial guess on P, e.g. $P(d = 0|x) = 0.99 \; \forall x$

Loop over $K = 1, 2, \ldots$:

1. MLE as-if "iterated P^{K-1}", i.e. $\tilde{P} = \Psi(P^{K-1}; \theta^{K-1})$ was solution $\hat{\theta}^K = \arg\max_\theta \frac{1}{N} \sum_{i=1}^N 1(d_i = 0)\tilde{P}(d_i = 0|x_i) + 1(d_i = 1)\tilde{P}(d_i = 1|x_i)$
2. Make policy iteration: $P^K = \Psi(P^{K-1}; \theta^K)$ with $\theta^K = \hat{\theta}^K$

Pseudo likelihood since we do MLE as-if ("wrong") solution is the true solution

Can show NPL-estimate converges to MLE by NFXP.

Your time to shine!